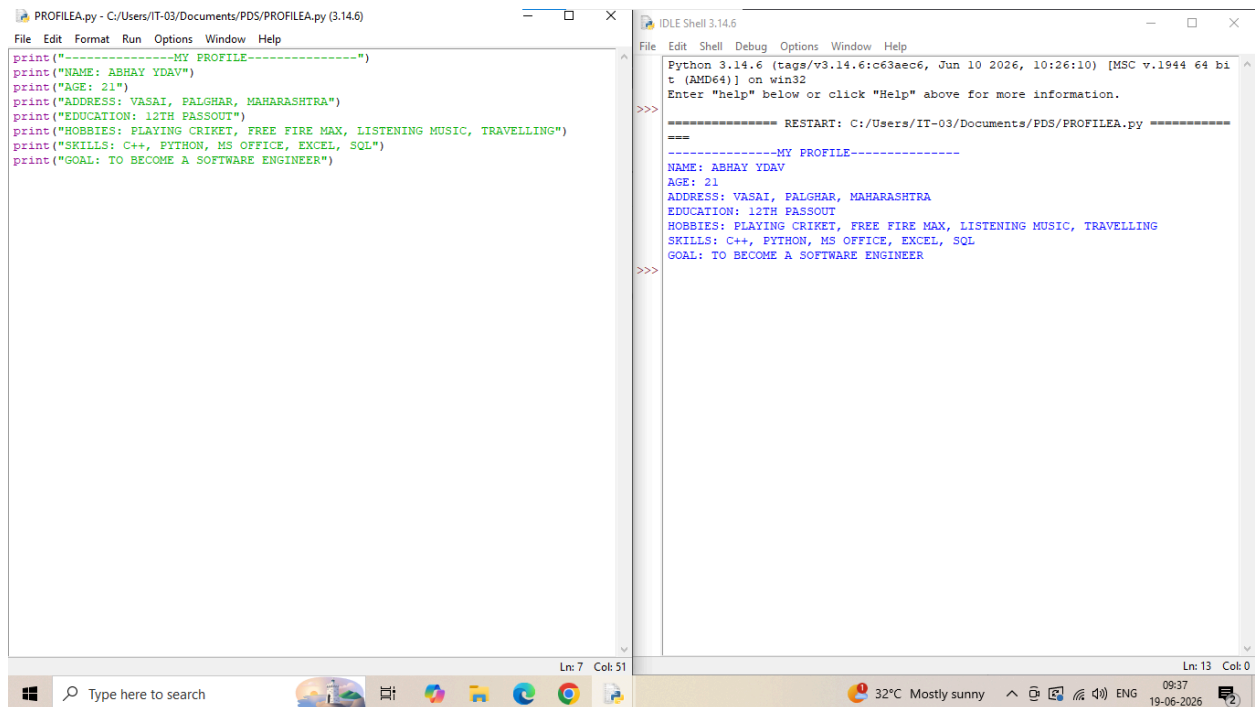


MVLU COLLEGE

1a. Write a Python code to print your profile.

Sol':



The screenshot displays a Python IDE with two windows. The left window, titled 'PROFILE.py - C:/Users/IT-03/Documents/PDS/PROFILE.py (3.14.6)', contains the following Python code:

```
print("-----MY PROFILE-----")
print("NAME: ABHAY YDAV")
print("AGE: 21")
print("ADDRESS: VASAI, PALGHAR, MAHARASHTRA")
print("EDUCATION: 12TH PASSOUT")
print("HOBBIES: PLAYING CRICKET, FREE FIRE MAX, LISTENING MUSIC, TRAVELLING")
print("SKILLS: C++, PYTHON, MS OFFICE, EXCEL, SQL")
print("GOAL: TO BECOME A SOFTWARE ENGINEER")
```

The right window, titled 'IDLE Shell 3.14.6', shows the output of the script after execution:

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-03/Documents/PDS/PROFILE.py =====
-----MY PROFILE-----
NAME: ABHAY YDAV
AGE: 21
ADDRESS: VASAI, PALGHAR, MAHARASHTRA
EDUCATION: 12TH PASSOUT
HOBBIES: PLAYING CRICKET, FREE FIRE MAX, LISTENING MUSIC, TRAVELLING
SKILLS: C++, PYTHON, MS OFFICE, EXCEL, SQL
GOAL: TO BECOME A SOFTWARE ENGINEER
>>>
```

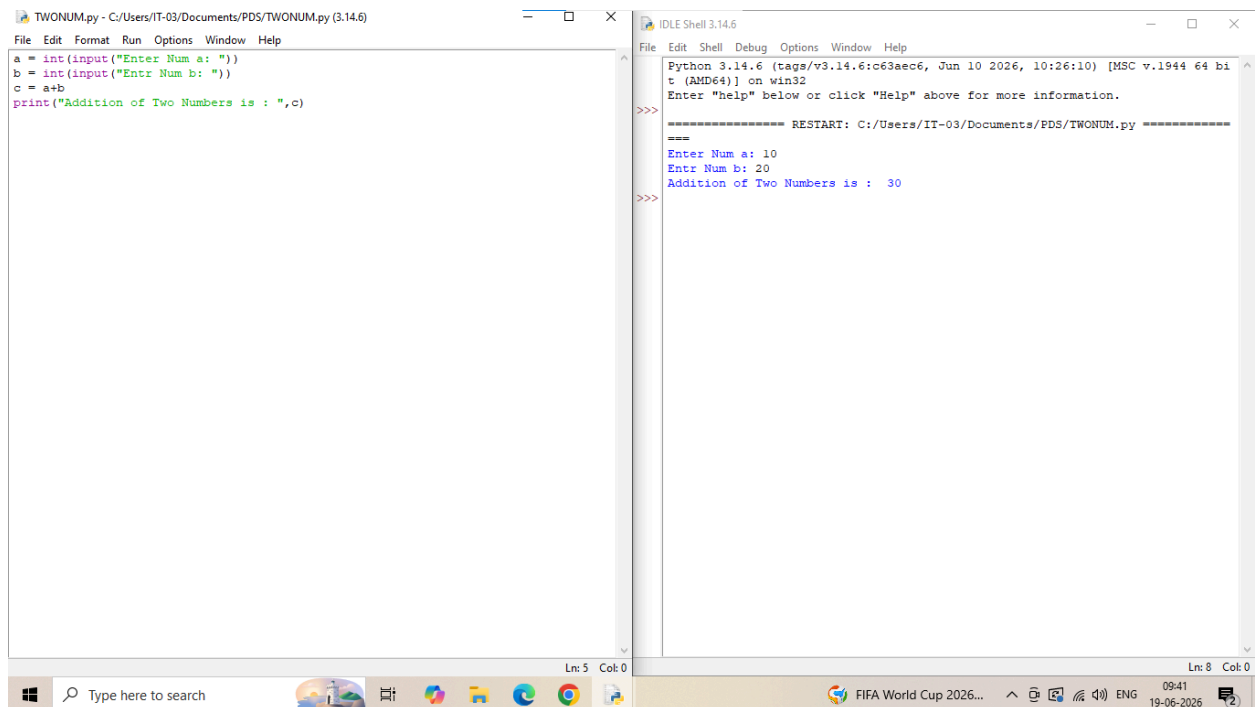
The Windows taskbar at the bottom shows the system clock as 09:37 on 19-06-2026, with a temperature of 32°C and weather 'Mostly sunny'.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1b. Write a Python code to print the addition of two numbers.

Sol':



The image shows a screenshot of a Python IDE with two windows. The left window, titled 'TWONUM.py - C:/Users/IT-03/Documents/PDS/TWONUM.py (3.14.6)', contains the following Python code:

```
a = int(input("Enter Num a: "))
b = int(input("Enter Num b: "))
c = a+b
print("Addition of Two Numbers is : ",c)
```

The right window, titled 'IDLE Shell 3.14.6', shows the execution output after running the script. It displays the prompts for input, the entered values (10 and 20), and the resulting sum (30).

```
Python 3.14.6 (tags/v3.14.6:c63aee6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-03/Documents/PDS/TWONUM.py =====
>>>
Enter Num a: 10
Enter Num b: 20
Addition of Two Numbers is : 30
>>>
```

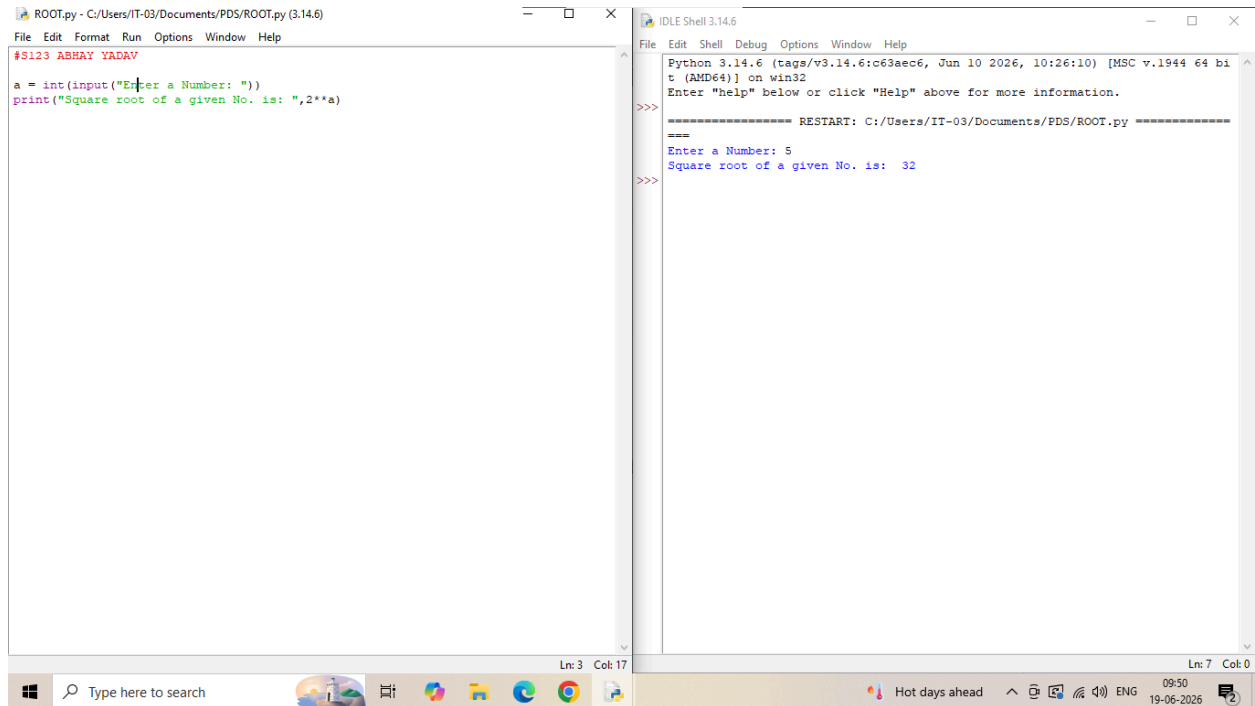
The Windows taskbar at the bottom shows the search bar, taskbar icons, and system tray with the date 19-06-2026 and time 09:41.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1c. Write a Python code to print the square root of a number.

Sol':



The screenshot displays two windows from a Windows operating system. The left window is a Python script editor titled 'ROOT.py - C:/Users/IT-03/Documents/PDS/ROOT.py (3.14.6)'. It contains the following code:

```
#S123 ABHAY YADAV  
  
a = int(input("Enter a Number: "))  
print("Square root of a given No. is: ",2**a)
```

The right window is an IDLE Shell titled 'IDLE Shell 3.14.6'. It shows the execution of the script. The output is as follows:

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi  
t (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/IT-03/Documents/PDS/ROOT.py =====  
Enter a Number: 5  
Square root of a given No. is: 32  
>>>
```

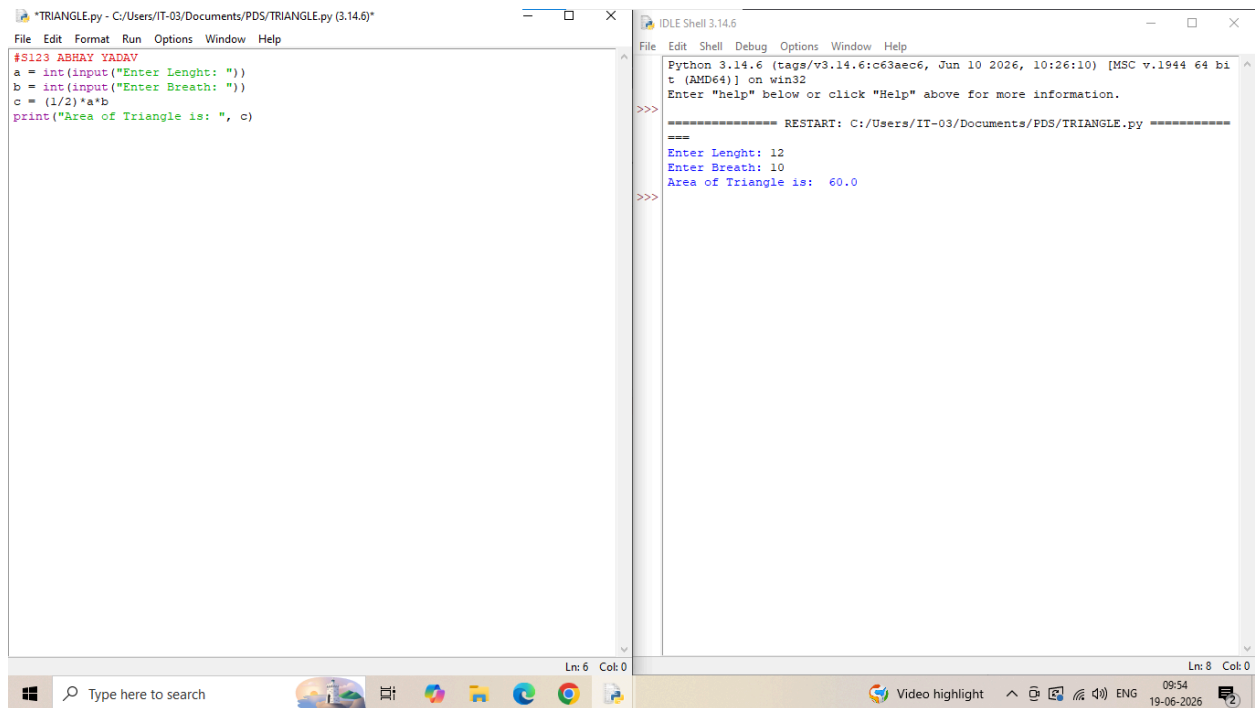
The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system tray on the right indicates the time as 09:50 on 19-06-2026, with the language set to ENG.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1c. Write a Python code to calculate the area of Triangle.

Sol':



The screenshot displays a Python IDE with two windows. The left window, titled "TRIANGLE.py - C:/Users/IT-03/Documents/PDS/TRIANGLE.py (3.14.6)", contains the following Python code:

```
#S123 ABHAY YADAV
a = int(input("Enter Length: "))
b = int(input("Enter Breath: "))
c = (1/2)*a*b
print("Area of Triangle is: ", c)
```

The right window, titled "IDLE Shell 3.14.6", shows the execution output:

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-03/Documents/PDS/TRIANGLE.py =====
>>>
Enter Length: 12
Enter Breath: 10
Area of Triangle is: 60.0
>>>
```

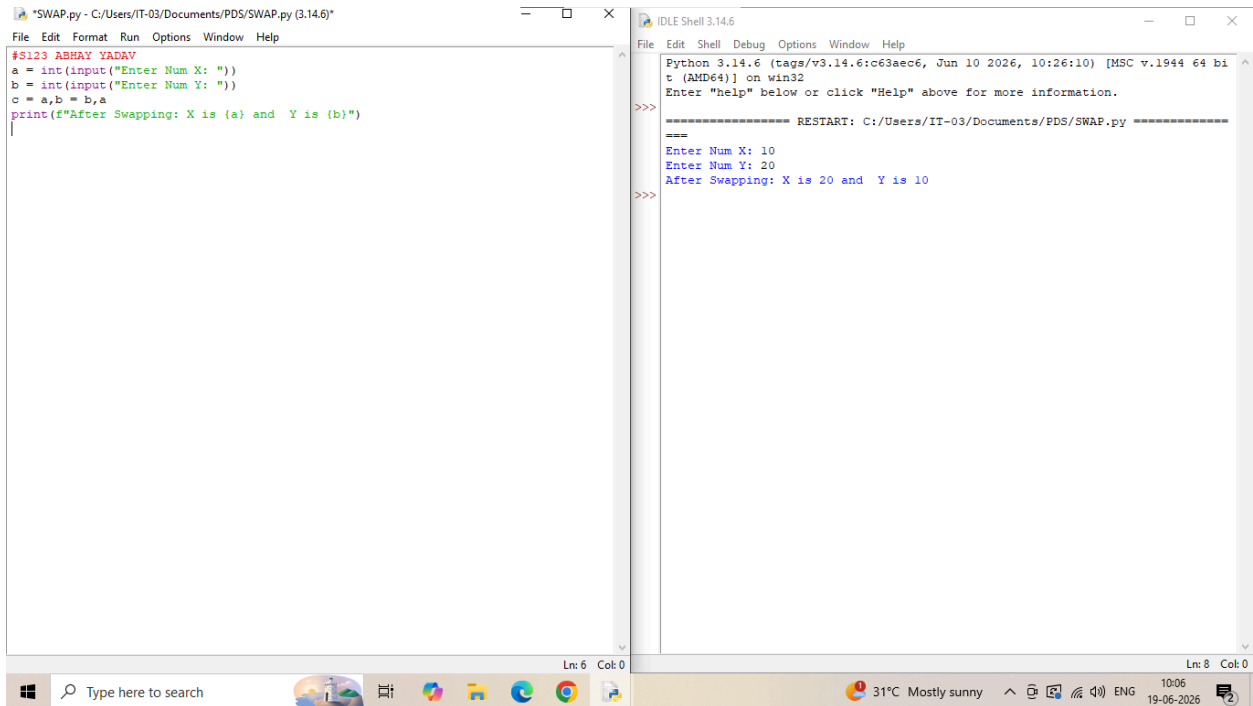
The Windows taskbar at the bottom shows the system clock as 09:54 on 19-06-2026.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1d. Write a Python code to swap two variables.

Sol':



The screenshot displays a Python IDE with two windows. The left window, titled "SWAP.py", contains the following code:

```
#S123 ABHAY YADAV
a = int(input("Enter Num X: "))
b = int(input("Enter Num Y: "))
c = a, b = b, a
print(f"After Swapping: X is {a} and Y is {b}")
```

The right window, titled "IDLE Shell 3.14.6", shows the execution output:

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-03/Documents/PDS/SWAP.py =====
>>>
Enter Num X: 10
Enter Num Y: 20
After Swapping: X is 20 and Y is 10
>>>
```

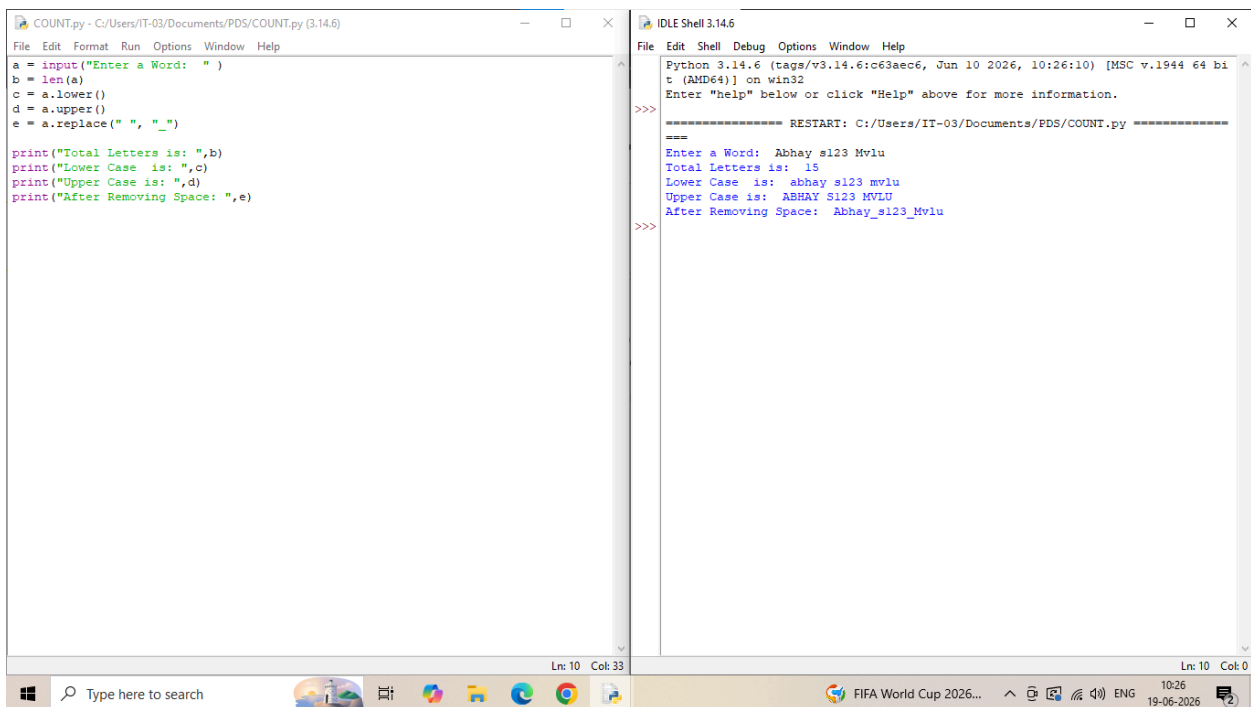
The Windows taskbar at the bottom shows the date as 19-06-2026 and the time as 10:06.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1e. Write a Python program that takes a sentence as input and counts the number of words and characters, converts the sentence to lowercase and uppercase, and replaces all spaces with underscores (_).

Sol':



The screenshot shows a Python IDE with two windows. The left window, titled 'COUNT.py - C:/Users/IT-03/Documents/PDS/COUNT.py (3.14.6)', contains the following Python code:

```
File Edit Format Run Options Window Help
a = input("Enter a Word: ")
b = len(a)
c = a.lower()
d = a.upper()
e = a.replace(" ", "_")

print("Total Letters is: ",b)
print("Lower Case is: ",c)
print("Upper Case is: ",d)
print("After Removing Space: ",e)
```

The right window, titled 'IDLE Shell 3.14.6', shows the output of the program after running it with the input 'Abhay s123 Mvlu':

```
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-03/Documents/PDS/COUNT.py =====
>>>
Enter a Word: Abhay s123 Mvlu
Total Letters is: 15
Lower Case is: abhay s123 mvlu
Upper Case is: ABHAY S123 MVLU
After Removing Space: Abhay_s123_Mvlu
>>>
```

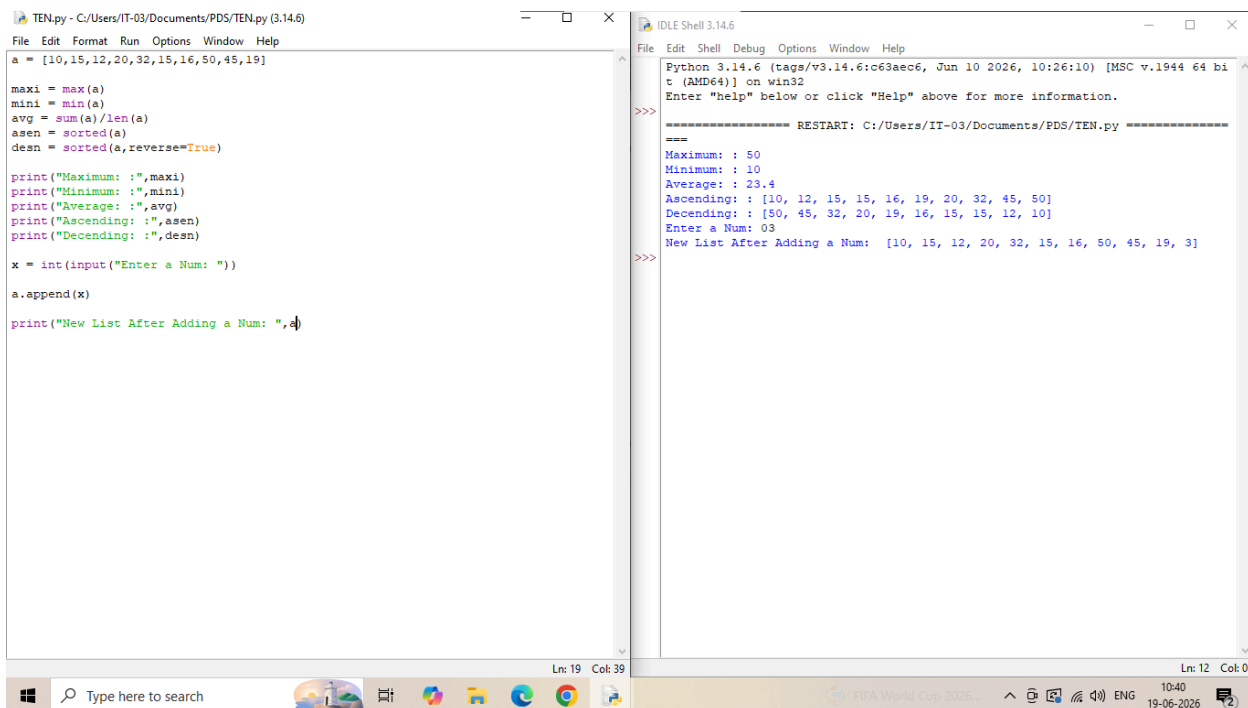
The Windows taskbar at the bottom shows the search bar, taskbar icons, and system tray with the date 19-06-2026 and time 10:26.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1f. Create a list of 10 numbers. Write a program to find the maximum, minimum, and average, sort the list in ascending and descending order, and add a new number to the list and remove the first item.

Sol':



```
TEN.py - C:/Users/IT-03/Documents/PDS/TEN.py (3.14.6)
File Edit Format Run Options Window Help

a = [10,15,12,20,32,15,16,50,45,19]

maxi = max(a)
mini = min(a)
avg = sum(a)/len(a)
asen = sorted(a)
desn = sorted(a,reverse=True)

print("Maximum: ",maxi)
print("Minimum: ",mini)
print("Average: ",avg)
print("Ascending: ",asen)
print("Decending: ",desn)

x = int(input("Enter a Num: "))

a.append(x)

print("New List After Adding a Num: ",a)
```

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aee6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

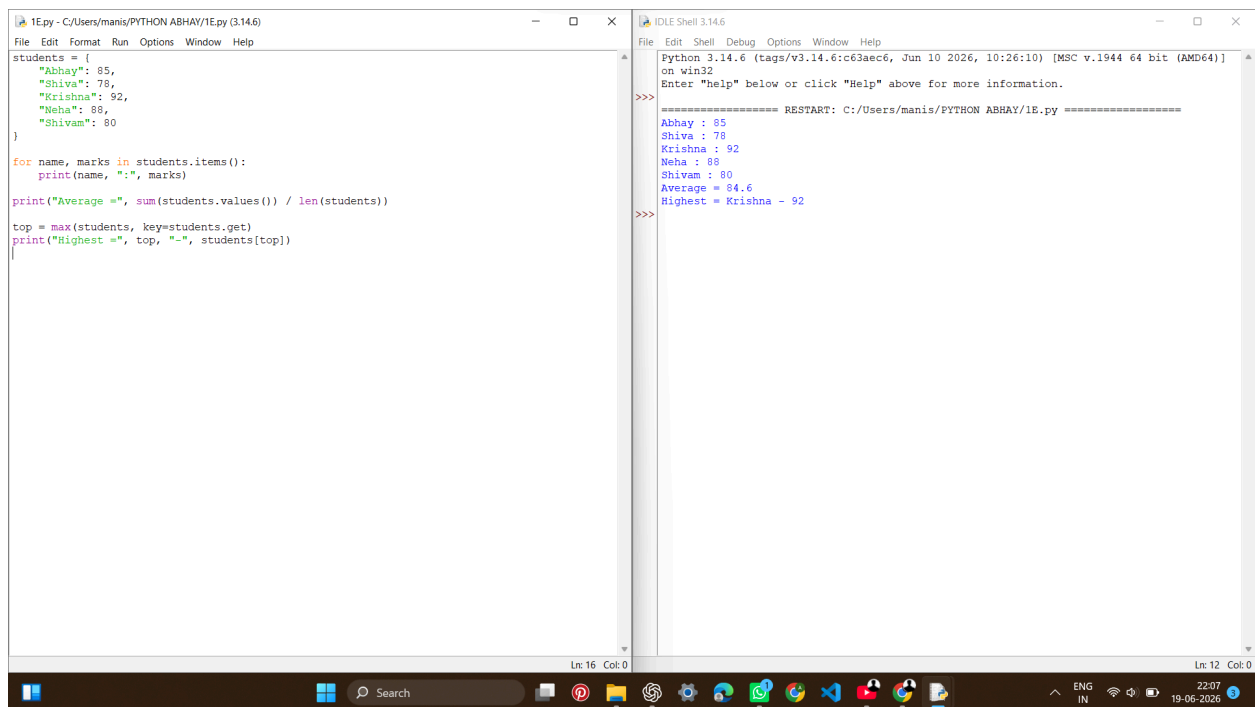
>>>
===== RESTART: C:/Users/IT-03/Documents/PDS/TEN.py =====
>>>
Maximum: : 50
Minimum: : 10
Average: : 23.4
Ascending: : [10, 12, 15, 15, 16, 19, 20, 32, 45, 50]
Decending: : [50, 45, 32, 20, 19, 16, 15, 15, 12, 10]
Enter a Num: 03
New List After Adding a Num: [10, 15, 12, 20, 32, 15, 16, 50, 45, 19, 3]
>>>
```

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1e. Store the names and marks of 5 students in a dictionary and then print each student's name and marks, calculate and print the class average, and find the student with the highest marks.

Sol':



The screenshot shows a Python IDE with two windows. The left window displays a Python script that defines a dictionary of student marks, iterates through it to print names and marks, calculates the class average, and identifies the student with the highest marks. The right window shows the output of the script after execution.

```
1E.py - C:/Users/manis/PYTHON ABHAY/1E.py (3.14.6)
File Edit Format Run Options Window Help
students = {
    "Abhay": 85,
    "Shiva": 78,
    "Krishna": 92,
    "Neha": 88,
    "Shivam": 80
}

for name, marks in students.items():
    print(name, ":", marks)

print("Average =", sum(students.values()) / len(students))
top = max(students, key=students.get)
print("Highest =", top, "-", students[top])

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

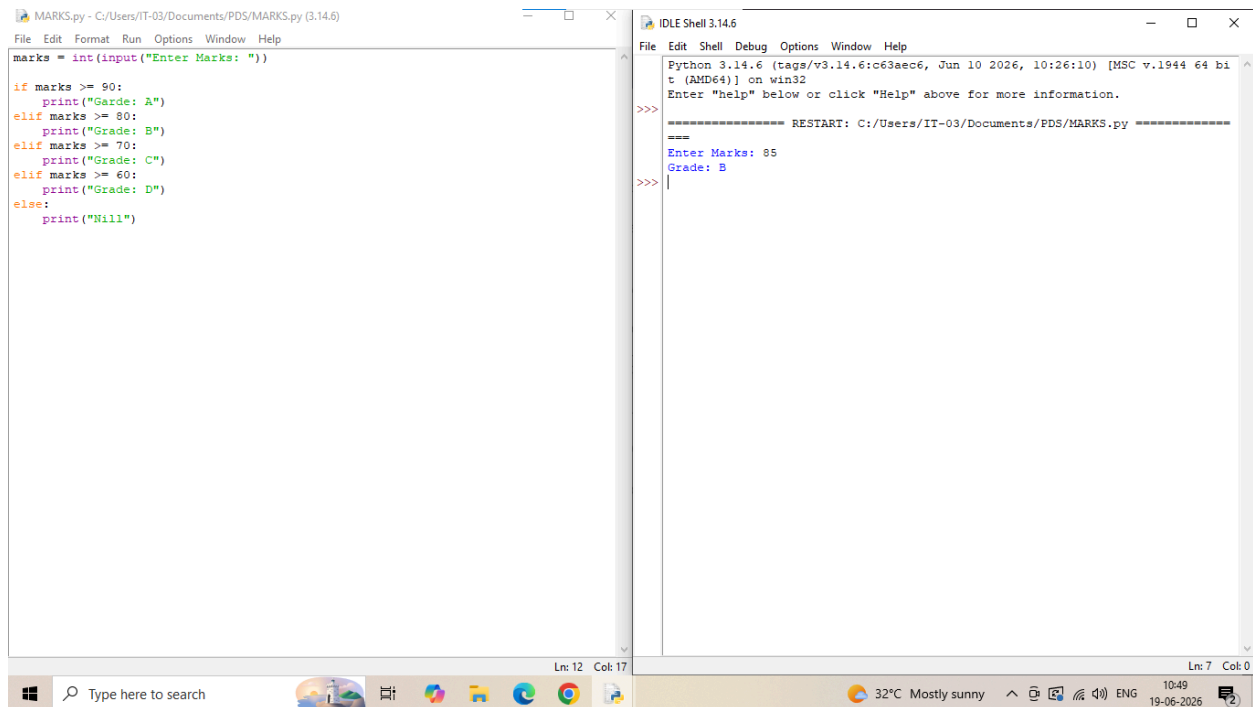
>>>
===== RESTART: C:/Users/manis/PYTHON ABHAY/1E.py =====
Abhay : 85
Shiva : 78
Krishna : 92
Neha : 88
Shivam : 80
Average = 84.6
Highest = Krishna - 92
>>>
```

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1g. Write a Python program that accepts marks as input and prints the grade as 90+ = A, 75–89 = B, 50–74 = C, below 50 = Fail.

Sol':



The screenshot shows a Python IDE with two windows. The left window, titled 'MARKS.py', contains the following code:

```
marks = int(input("Enter Marks: "))

if marks >= 90:
    print("Grade: A")
elif marks >= 80:
    print("Grade: B")
elif marks >= 70:
    print("Grade: C")
elif marks >= 60:
    print("Grade: D")
else:
    print("Fail")
```

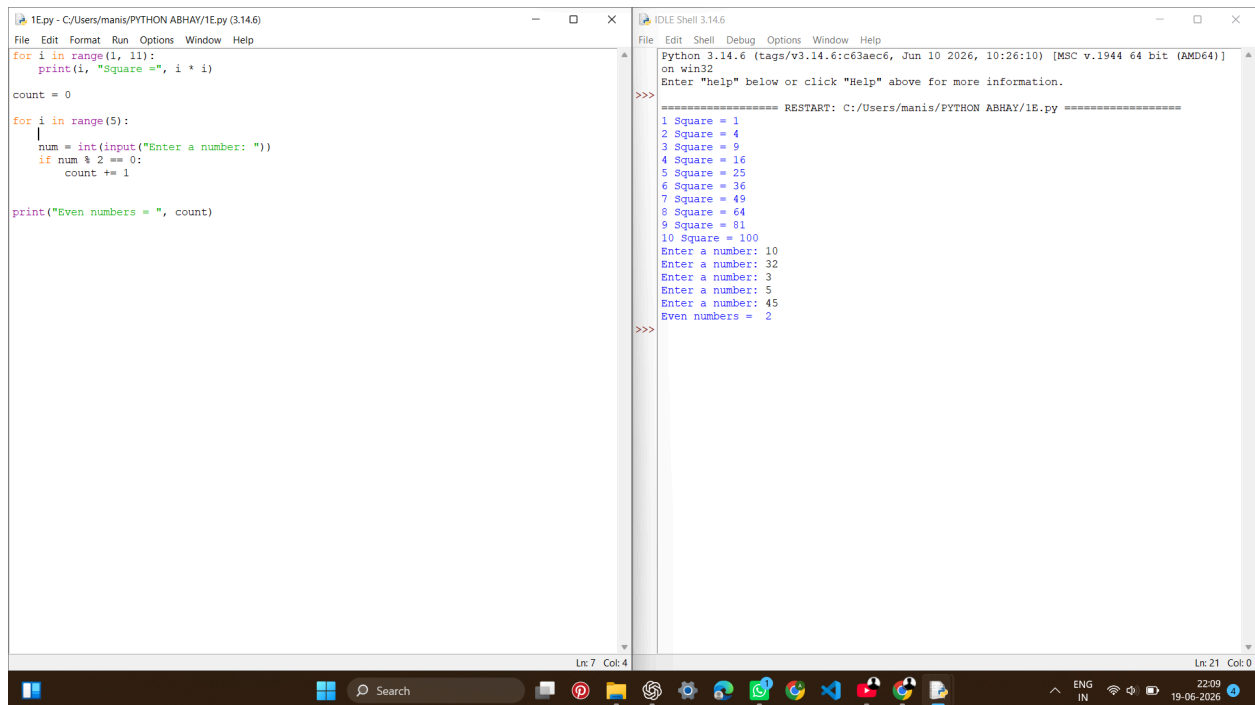
The right window, titled 'IDLE Shell 3.14.6', shows the execution of the program. It displays the prompt 'Enter Marks: 85' and the output 'Grade: B'.

NAME: ABHAY YADAV
ROLL NO. S123

MVLU COLLEGE

1h. Write a Python program using a loop that prints the square of numbers from 1 to 10, accepts 5 numbers from the user and prints how many are even.

Sol':



```
1E.py - C:/Users/manis/PYTHON ABHAY/1E.py (3.14.6)
File Edit Format Run Options Window Help
for i in range(1, 11):
    print(i, "Square =", i * i)

count = 0

for i in range(5):
    num = int(input("Enter a number: "))
    if num % 2 == 0:
        count += 1

print("Even numbers = ", count)
```

```
IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/manis/PYTHON ABHAY/1E.py =====
1 Square = 1
2 Square = 4
3 Square = 9
4 Square = 16
5 Square = 25
6 Square = 36
7 Square = 49
8 Square = 64
9 Square = 81
10 Square = 100
Enter a number: 10
Enter a number: 32
Enter a number: 3
Enter a number: 5
Enter a number: 45
Even numbers = 2

>>>
```

NAME: ABHAY YADAV
ROLL NO. S123